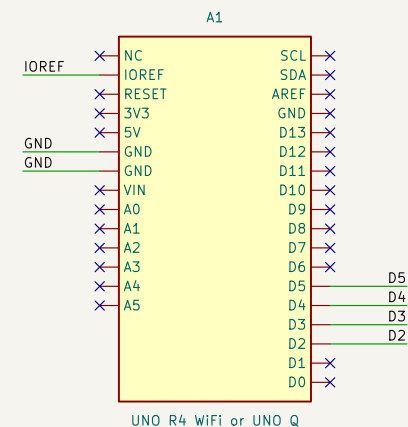
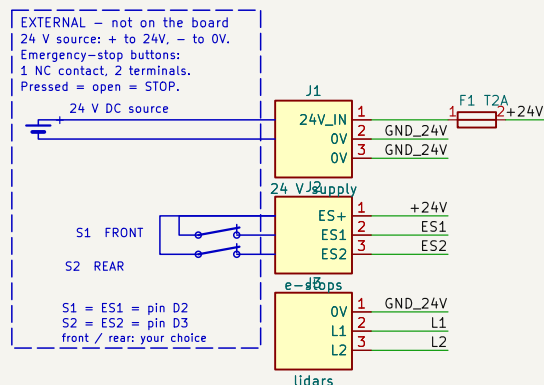
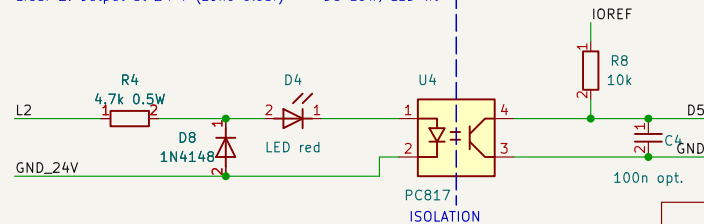


E-stop loops: 3.3 k $\rightarrow$ 6.3 mA (0.20 W at 28.8 V). Lidar inputs: 4.7 k $\rightarrow$ 4.4 mA (0.14 W). R1-R4 are 1/2 W (body 9 x 3.2 mm); pull-ups 1/4 W. No current (button pressed, zone occupied, wire cut, 24 V lost) $\rightarrow$ pin HIGH = STOP. F1 protects the 24 V wires going out to the buttons. GND_24V and GND are NOT connected: this is the isolation barrier. Pull-ups go to IOREF (5 V on UNO R4 WiFi, 3.3 V on UNO Q), never to 5V. C1-C4 are OPTIONAL: fitted, an opening shorter than about 1.5 ms is not seen by the Arduino, and every stop is seen about 1.6 ms later. E-stop 1: button released (contact closed) $\rightarrow$ D2 LOW, LED lit



Sheet: /
File: pc817_bench.kicad_sch

Id: 1/1